

Problem 65. Let ABC be an acute-angled triangle with $AB < AC$. Let D be the foot of the internal angle bisector from A to BC and let M be the midpoint of the line segment AD . Finally, let X be a point on the line segment BM such that $\angle MXA = \angle DAC$. Show that the lines AX and XC are perpendicular to each other.

Problem 66. Let $\lambda \geq 1$ be a real number and let n be a positive integer such that

$$\lfloor \lambda^{n+1} \rfloor, \lfloor \lambda^{n+2} \rfloor, \dots, \lfloor \lambda^{4n} \rfloor \text{ are all perfect squares,}$$

where $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

Prove that $\lfloor \lambda \rfloor$ is a perfect square.

Problem 67. Let x be a real number with $0 < x < 1$ and let $0.c_1c_2c_3\dots$ be the decimal expansion of x . Let $B(x)$ be the set of all substrings $c_ic_{i+1}\dots c_{i+5}$, where $i = 1, 2, \dots$, in the decimal expansion of x that consist of 6 consecutive digits.

For instance, $B(1/22) = \{045454, 454545, 545454\}$.

Find the minimum number of elements of $B(x)$ as x varies over the irrational numbers between 0 and 1 (i.e. among all real numbers in that range whose decimal expansion is neither finite nor eventually periodic).

Problem 68. In a certain club, some pairs of members are friends. Given $k \geq 3$, we say that a club is k -good if every group of k members can be seated around a round table such that every two neighbours are friends.

- (a) Prove that if a club is 6-good then it is 7-good.
- (b) Give an example of a club which is 9-good but not 10-good.